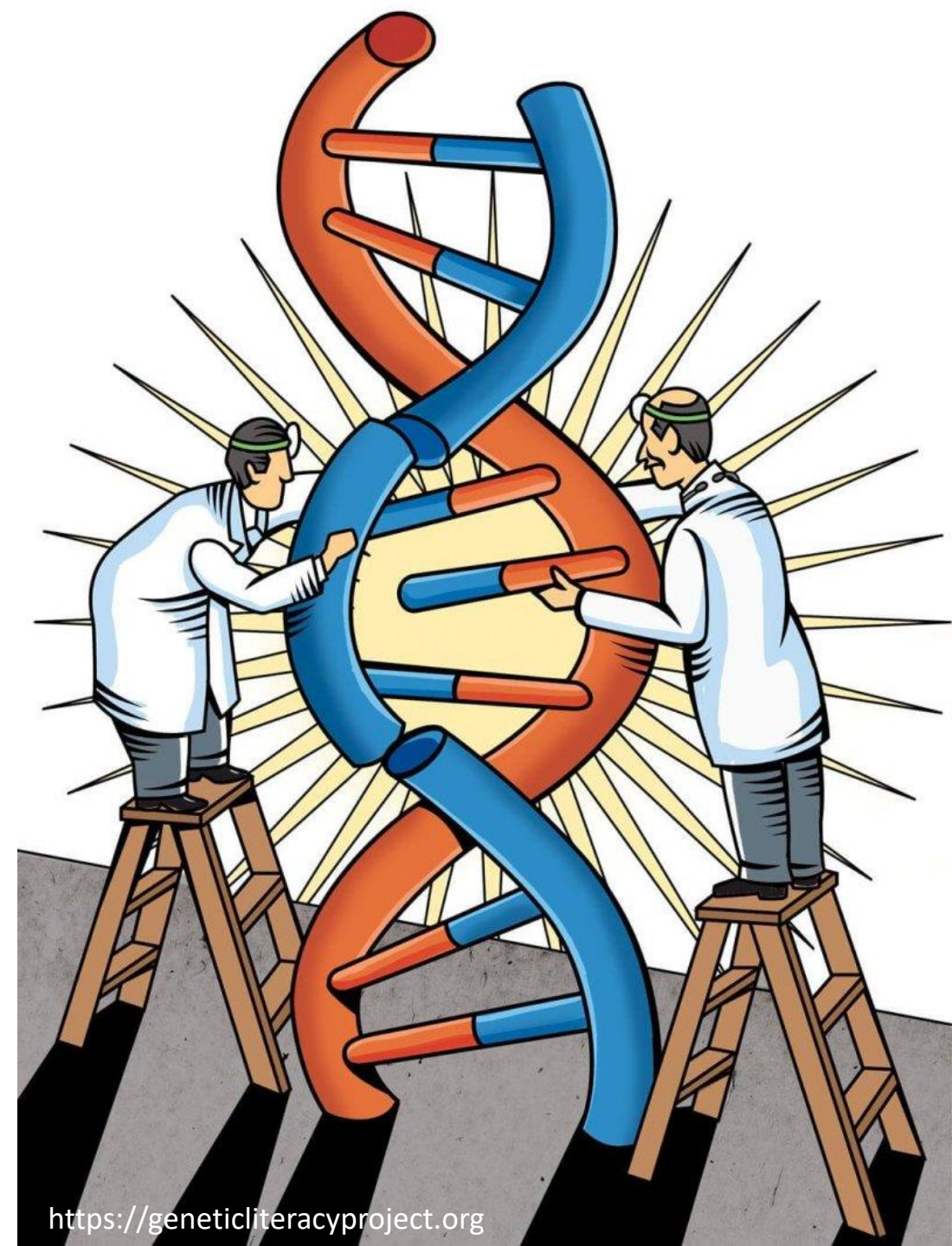


CRISPR in action

practice and theory of genome engineering

Day 1
August 17, 2026

Sibylle Vonesch
Laboratory for Genome Editing and
Systems Genetics



Trainers



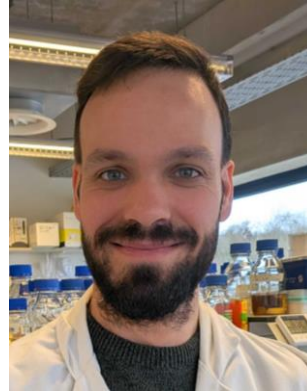
Ruben Vazquez Uribe

ruben.vazquezuribe@kuleuven.be



Sibylle Vonesch

sibylle.vonesch@kuleuven.be



Simon Fiesack
(Day 1 & 2)



Jana Nysten
(Day 1 & 2)



Siebe Thys
(Day 3 & 4)



Vlad Batagui
(Day 3 & 5)

Organizational matters

Please arrive on time and check schedule

Course rooms:

- LAND 00.207 – theory and *in silico* modules
- LAND 00.60 – wetlab

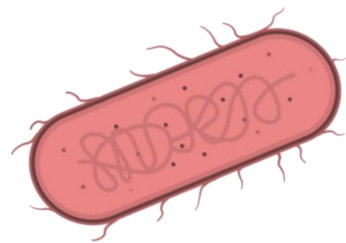
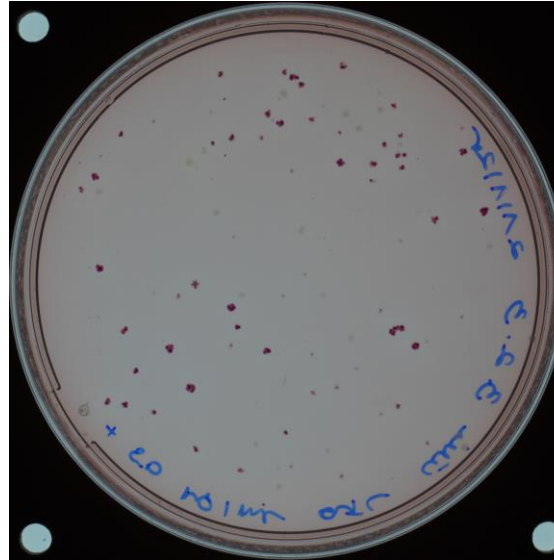
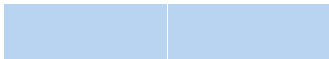
Breaks – LAND 00.208

Lab safety, VGM

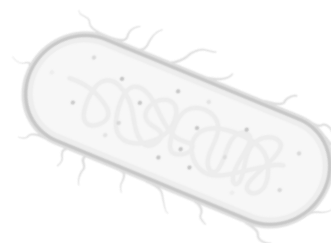
Editing experiments performed in the course

Experiment 1

Whole-gene deletion



galK present and functional



galK deleted or inactivated

Experiment 2

Precise knockout via PTC
(premature termination codon)

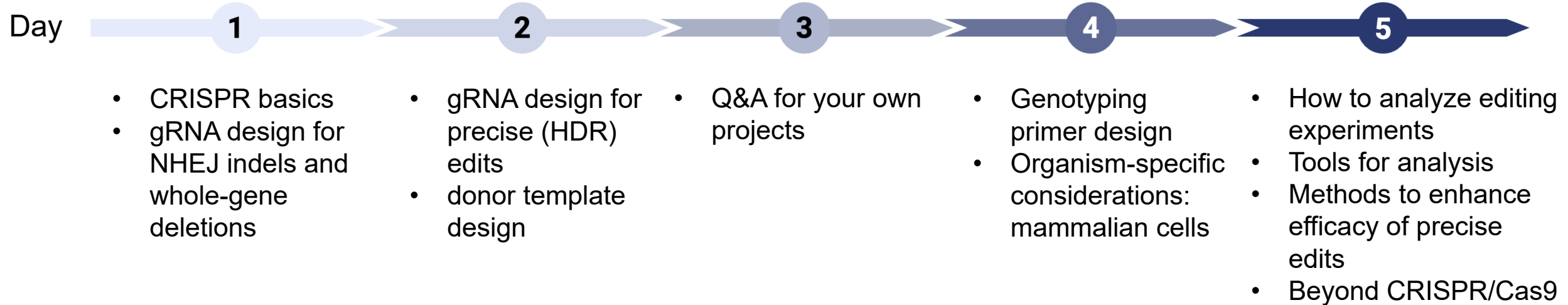
```
CTGCCGCTGGACGGCGCACAAATCGCGCTTAACGGTCAGGAAGCAGAAAACAGTTT  
GACGGCGACCTGCCGCGTGTTTAGCGCGAATTGCCAGTCCTTCGTCTTTGGTCAAA
```

148	150	152	154	156	158	160	162	164										
L	P	L	D	G	A	Q	I	A	L	N	G	Q	E	A	E	N	Q	F

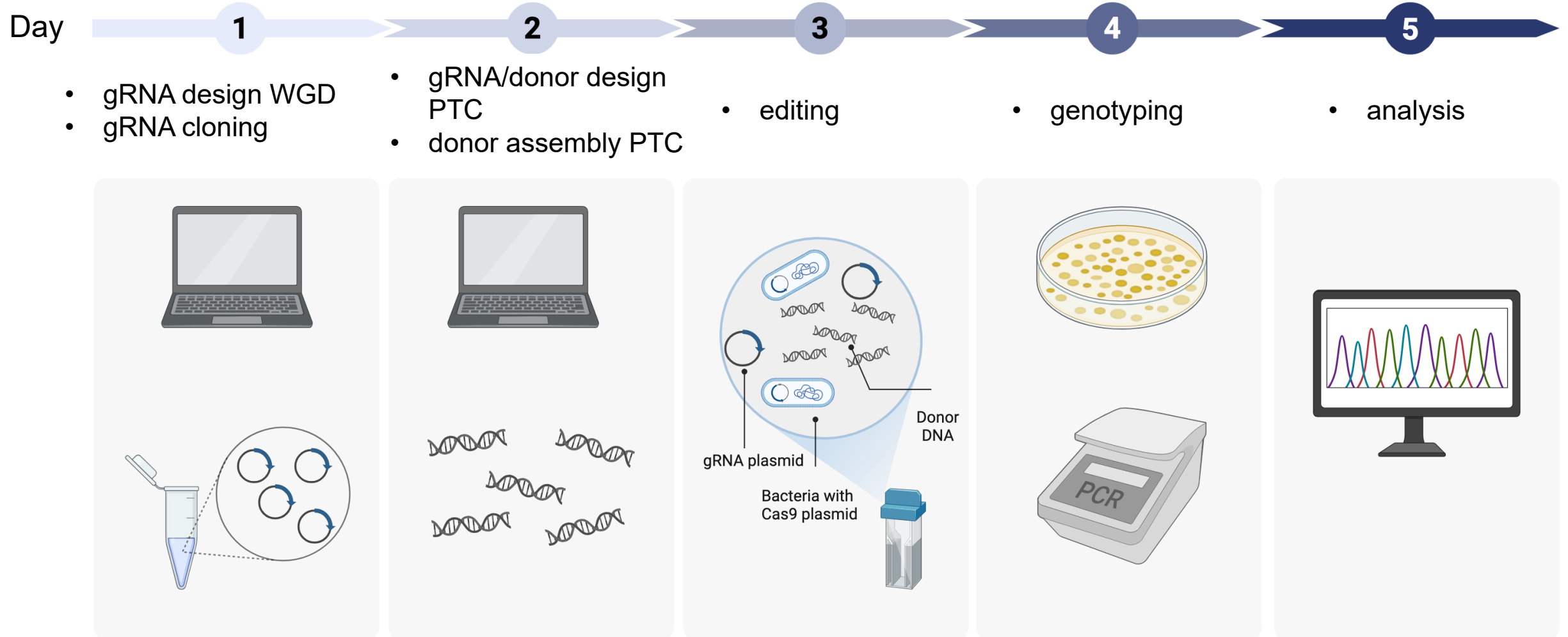
galK_coding

e.g. L156* (CTT to TAA)

Course overview – theory and drylab



Course overview - experiments



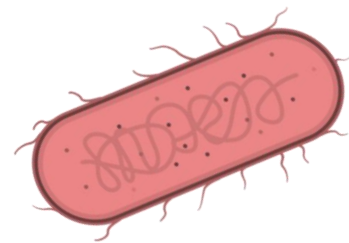
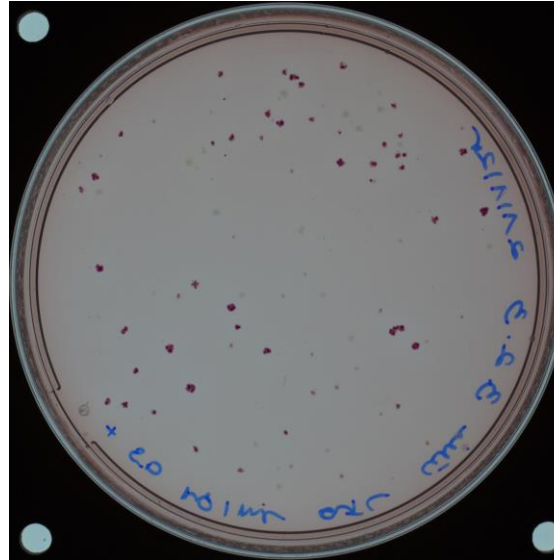
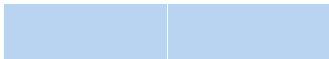
Group distribution

Background lab part – Day 1

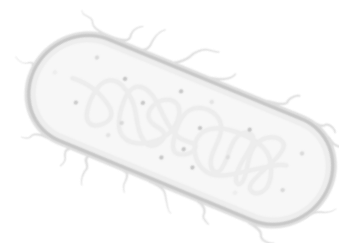
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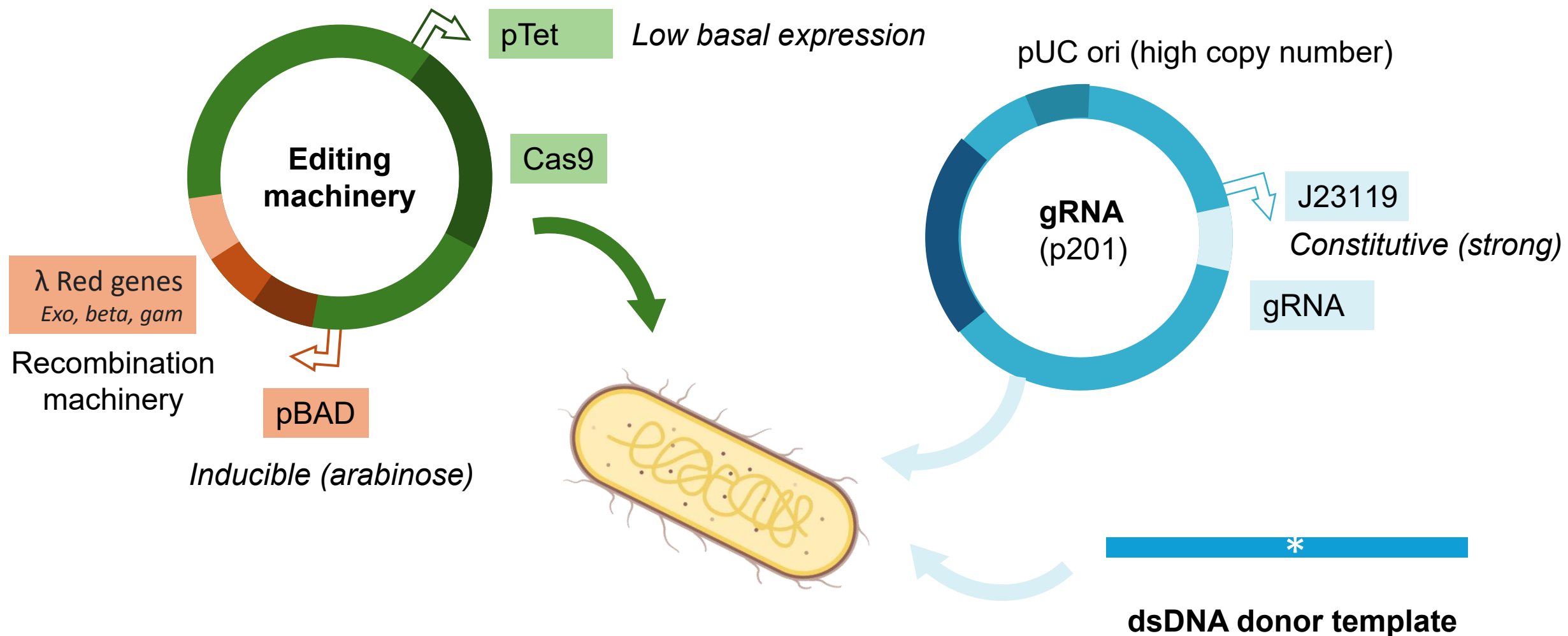
```
CTGCCGCTGGACGGCGCACAAATCGCGCTTAACGGTCAGGAAGCAGAAAACAGTTT  
GACGGCGACCTGCCGCGTGTTTAGCGCGAATTGCCAGTCCTTCGTCTTTGGTCAAA
```

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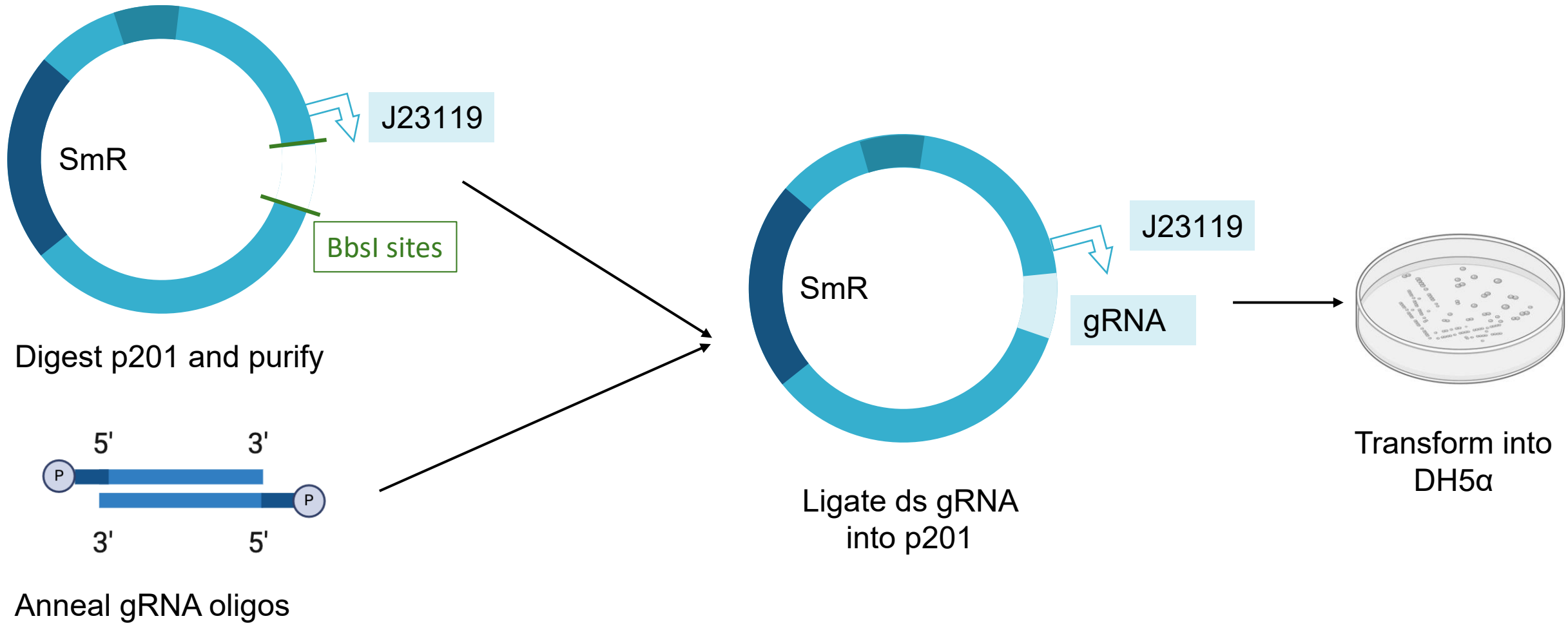
galK_coding

e.g. L156* (CTT to TAA)

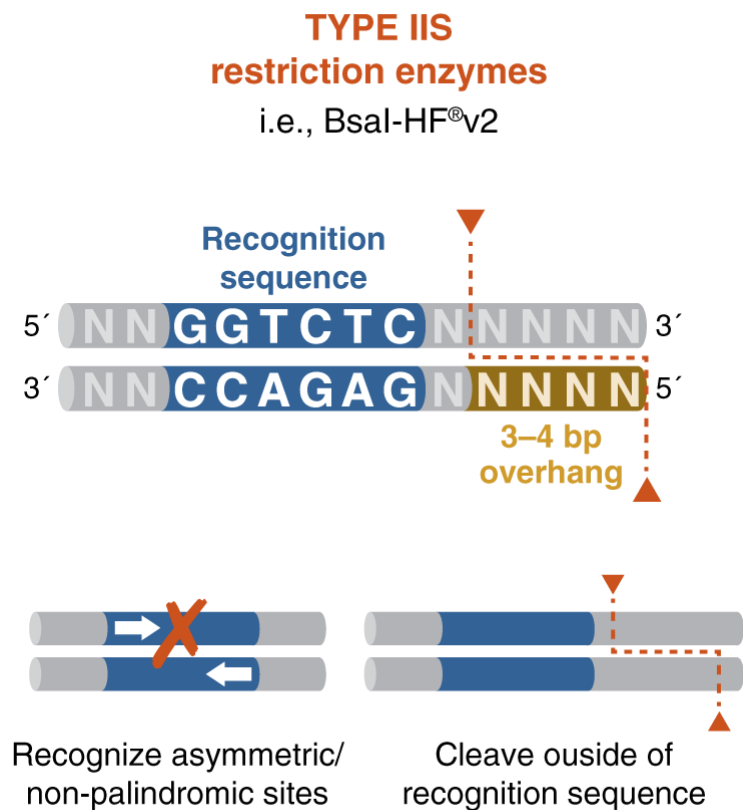
Editing system used in the course



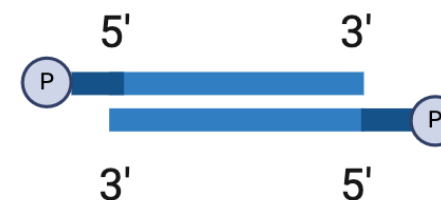
Experimental steps Day 1



gRNA cloning with Type IIS enzymes

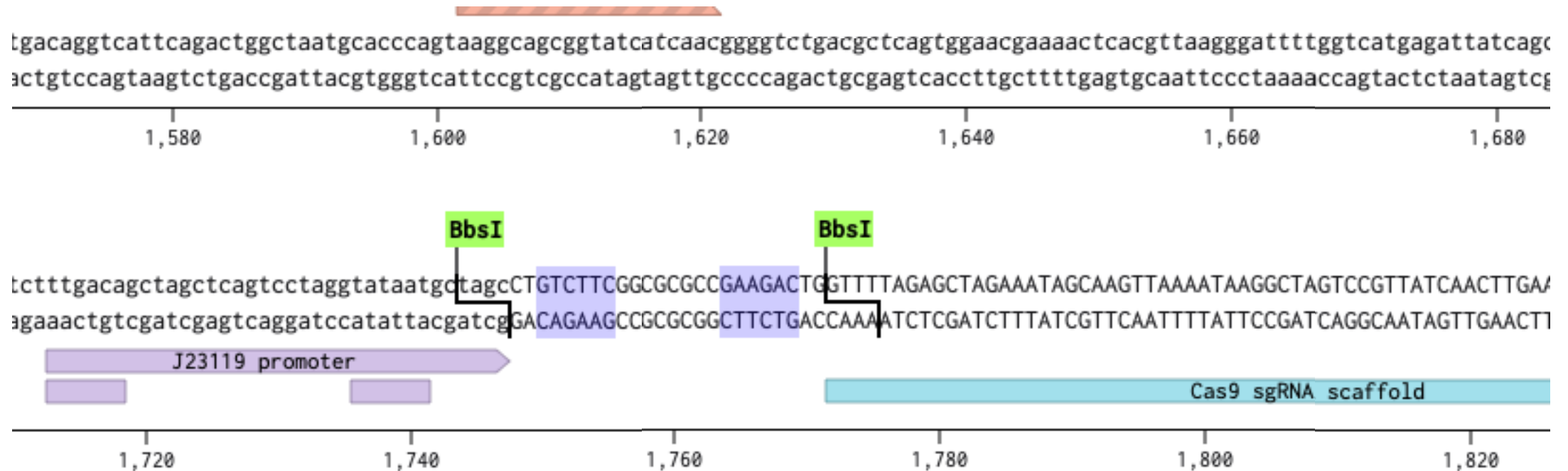


gRNA FWD and REV oligos with compatible overhangs



Digested backbone (p201) with compatible overhangs

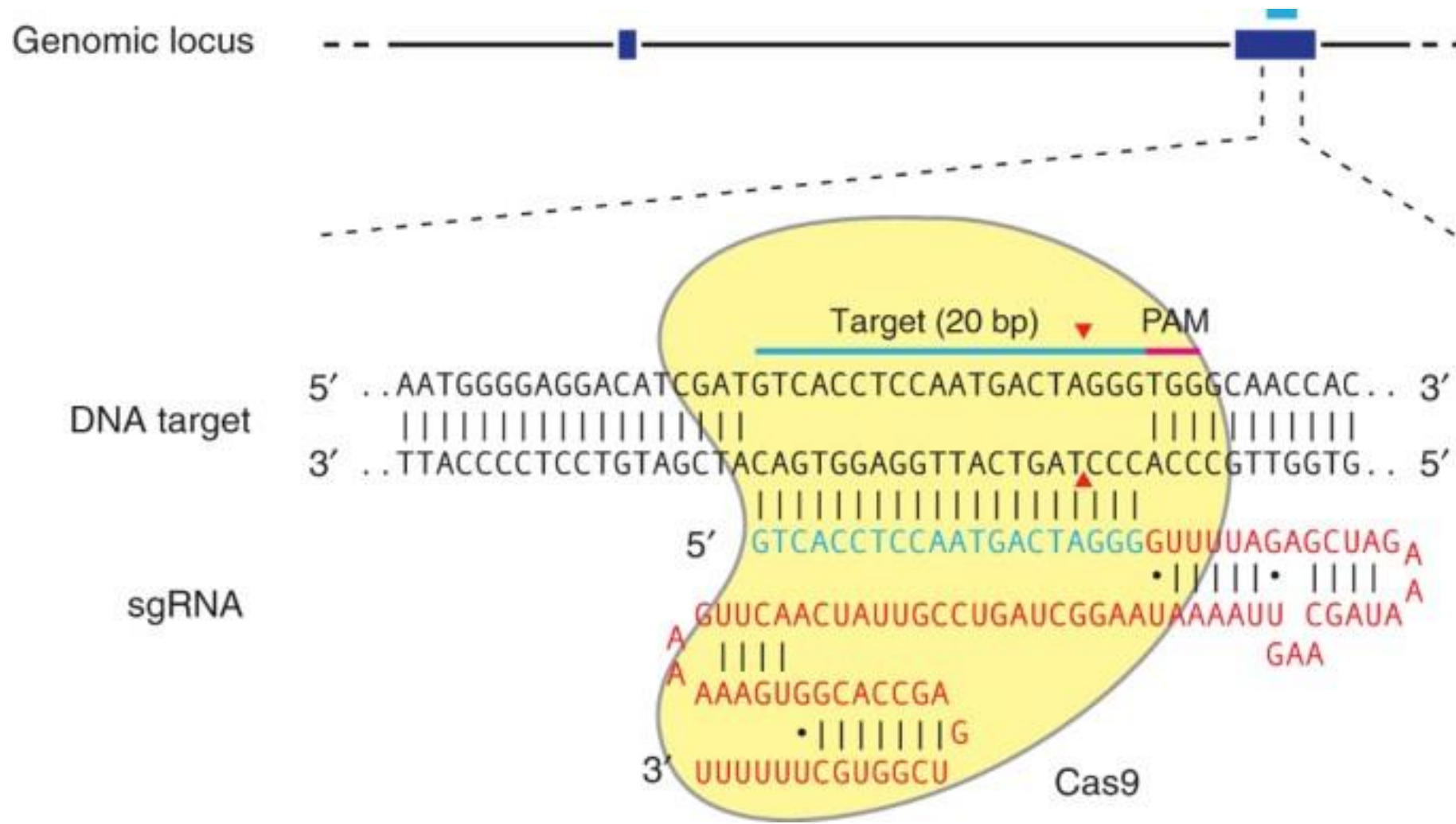
p201 entry vector – gRNA cloning site

[illegible]

p201: <https://benchling.com/s/seq-DQQIkiWn01ZzNBo7LPKs?m=sIm-mFnnggiqZE0R3zOuWHKa>

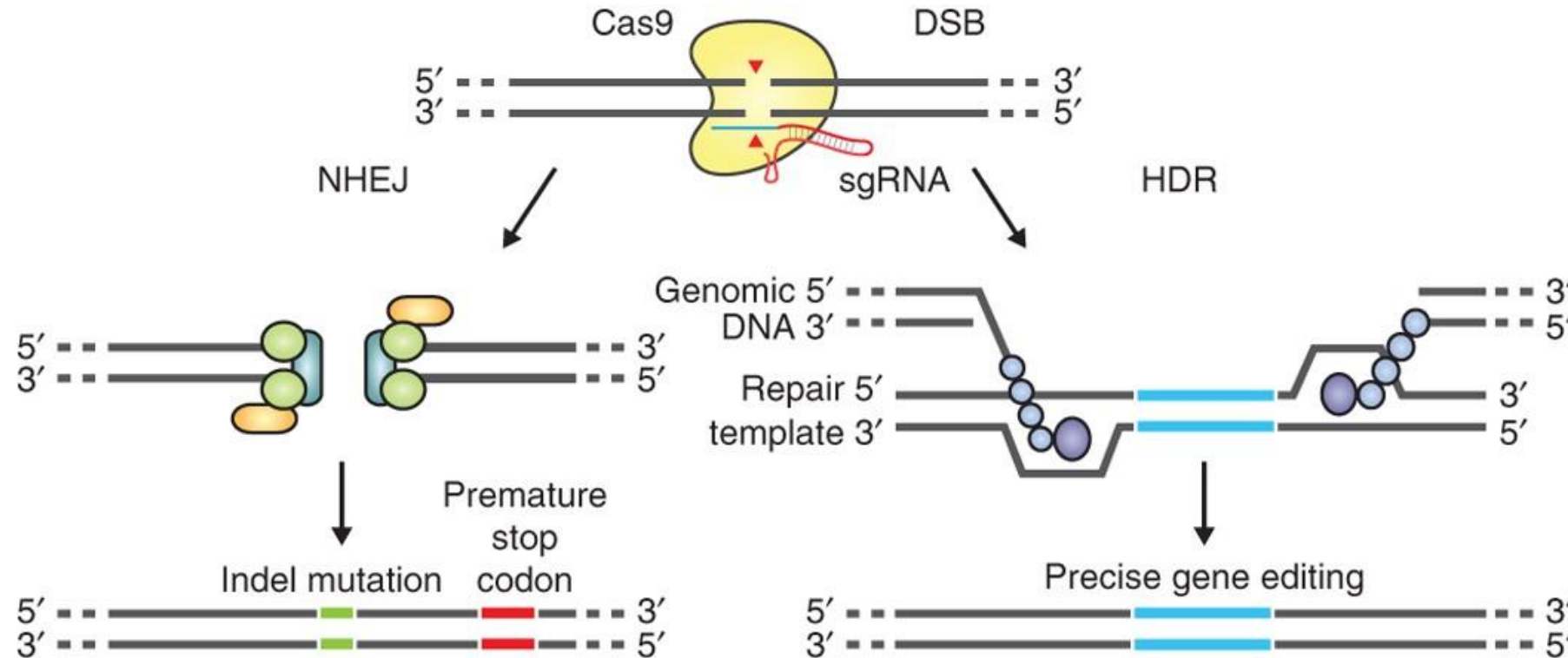
**gRNA design for NHEJ
frameshifts or deletion of entire
genes**

Components of the *S. pyogenes* CRISPR/Cas9 system



Genome modification using CRISPR/Cas

Double-strand break is lethal → repair by endogenous host pathways is required



Non-Homologous End Joining:

Indels → frameshift → premature stop
→ present in eukaryotes and only a few prokaryotes

Homology-Directed Repair

→ recombination with intact, homologous rescue DNA (= donor), introduction of designed sequences

gRNA design considerations for different CRISPR applications

LOCATION
SEQUENCE

Knockout (whole gene deletion, NHEJ indels)

LOCATION
SEQUENCE

CRISPRi/a

LOCATION
SEQUENCE

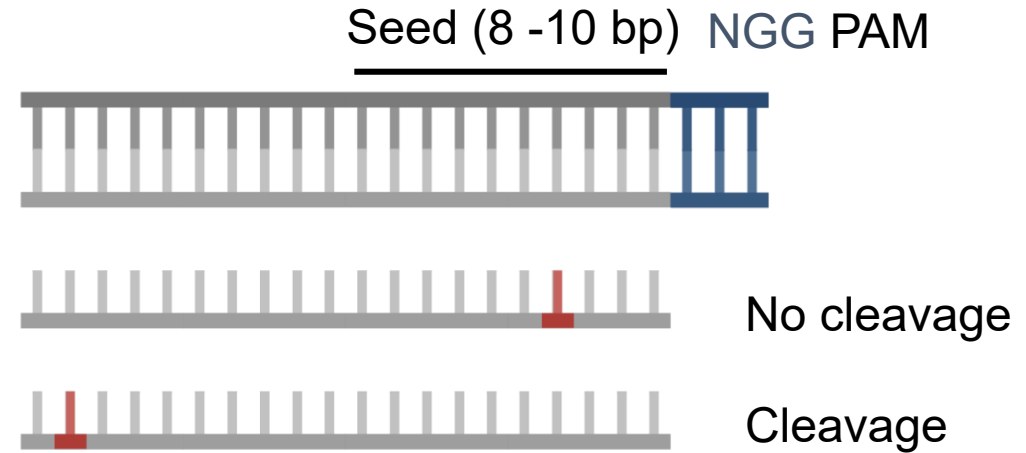
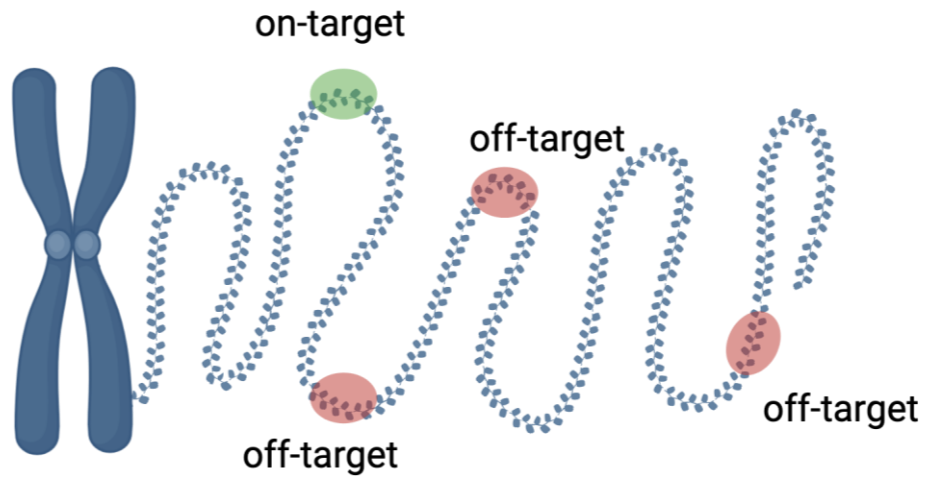
Precision genome editing (usually base/amino acid changes)

How to design gRNAs for NHEJ indels or whole-gene knockouts with *SpCas9*?



- Search for valid PAMs in your target gene (*SpCas9* cleavage site 3 nt upstream of NGG PAM)
- Choose best gRNA(s) – how?
 - Off-target activity
 - At least 3 mismatches to next best target, seed mismatches
 - Advanced models
 - On-target activity
 - GC content – 40-60% often optimal
 - Homopolymers – avoid T-stretches (>T4)
 - Advanced models with more features: secondary structure, specific nucleotides, chromatin state, ...
– integrated in Benchling, CHOPCHOP, CRISPOR
 - Location in gene
 - Avoid too close to start or end – alternative start codons, truncated but functional proteins
 - Exons

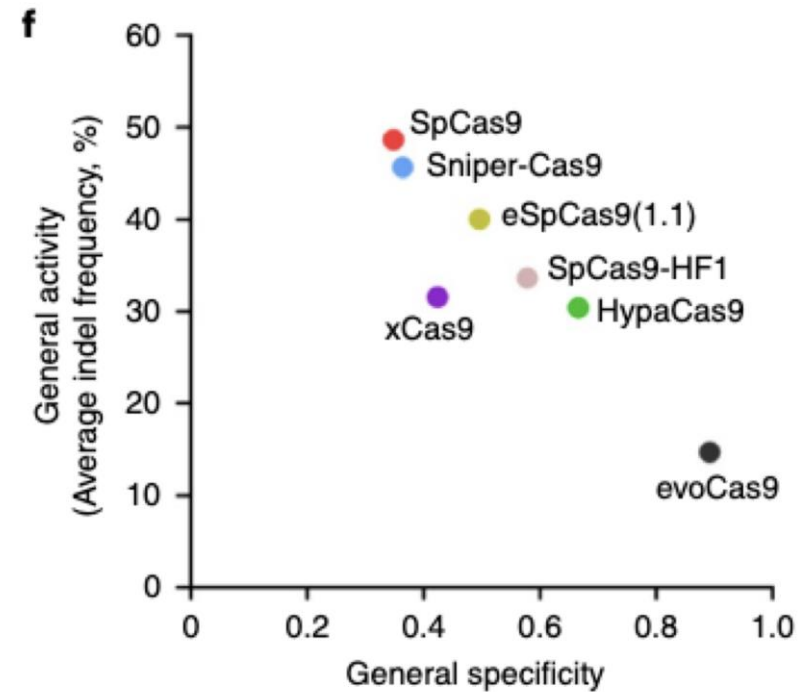
Off-target activity



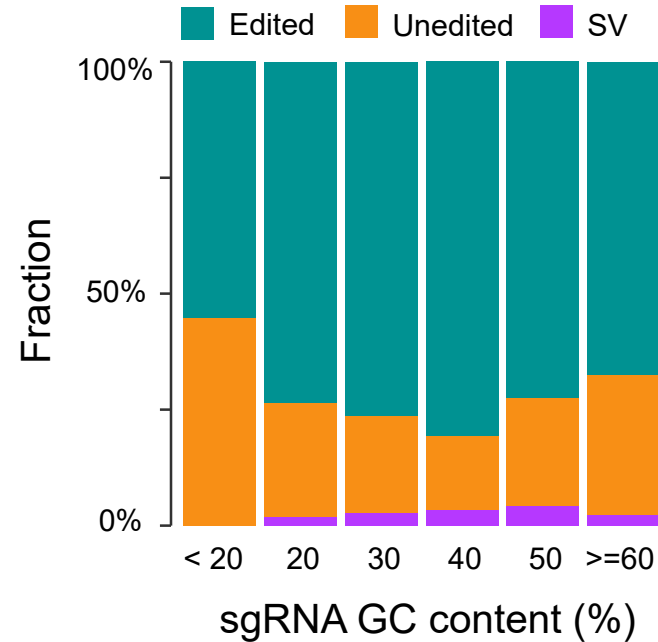
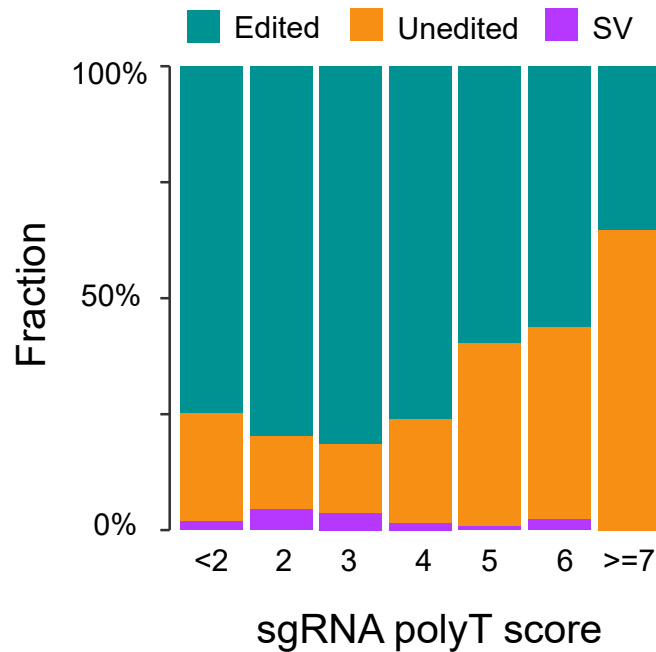
- Mismatches in seed region better at preventing off-target cleavage
- More repetitive/larger genomes – more off-targets
- Use design/prediction tools:
 - CasOFF-Finder: <http://www.rgenome.net/cas-offfinder/> - custom genomes, any nuclease (PAM)
 - CRISPOR: <https://crispor.gi.ucsc.edu> – many organisms and nucleases
 - CHOPCHOP: <https://chopchop.cbu.uib.no> – many organisms, add new species
- Aim for at least 3 mismatches to target, ideally at least one in seed
- **Make your mutation with 2 different gRNAs if possible**

Cas enzymes with improved specificity

Typically, a trade-off between efficiency and specificity

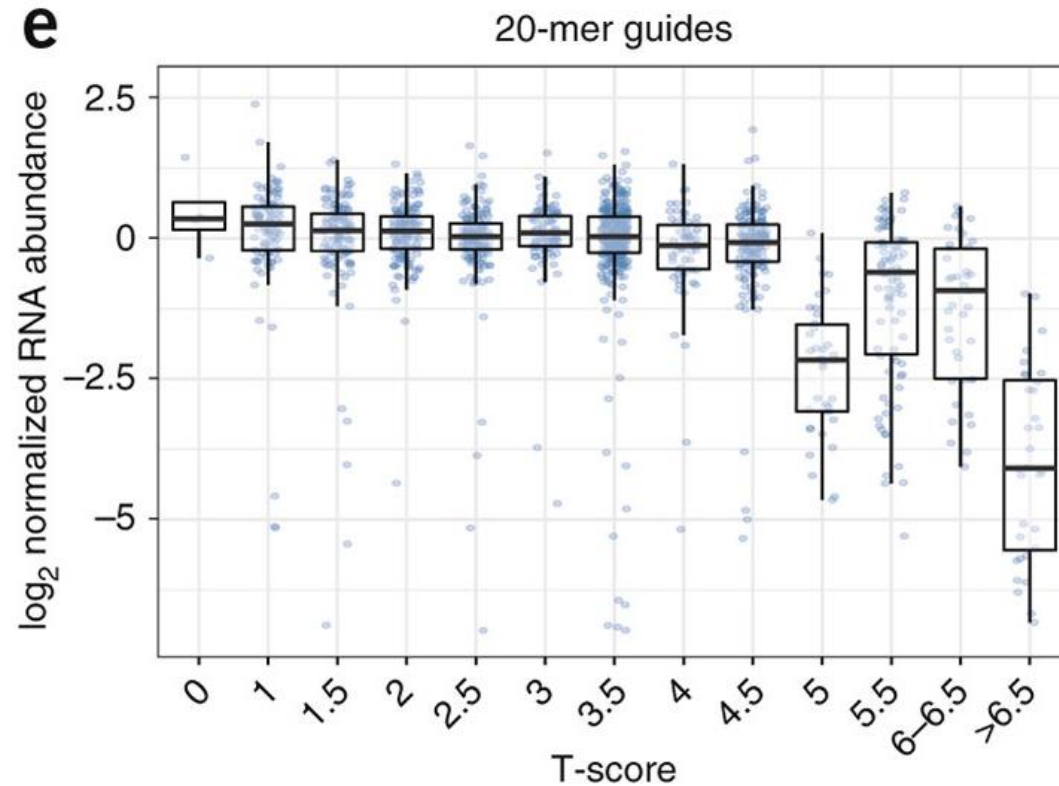


On-target activity: sequence features correlated with efficacy



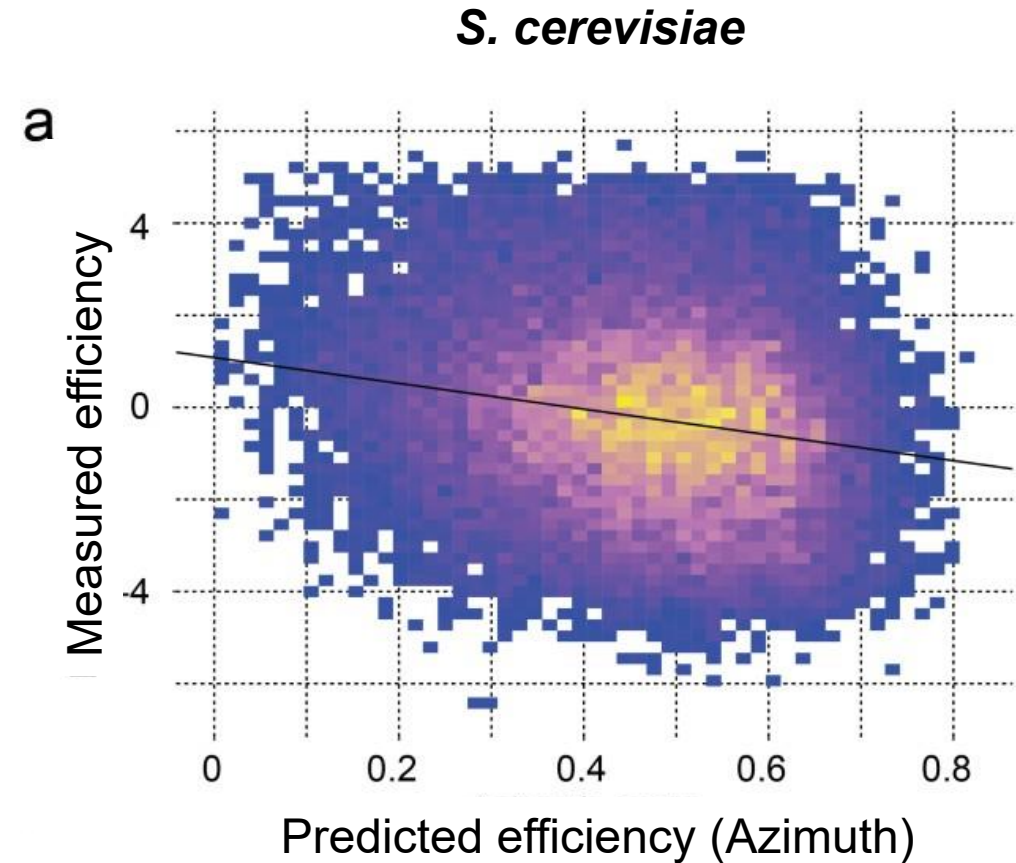
- 2688 different *SpCas9* genomic targets in *S. cerevisiae*
- Precision editing outcome measured via WGS

poly-T stretches in gRNA lead to reduced gRNA abundance



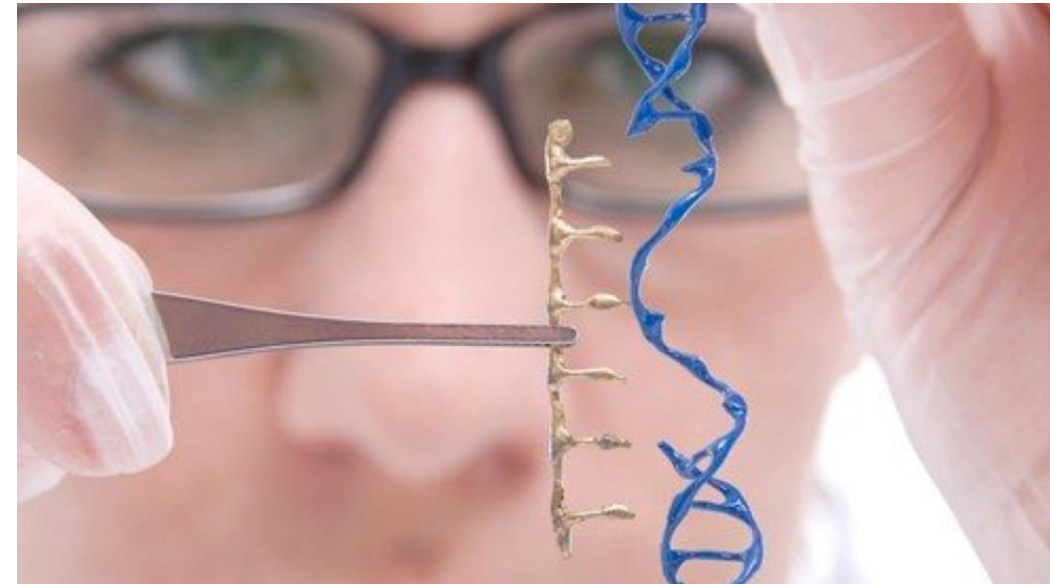
On-target activity: predictive models

- Cleavage efficiency models (e.g. Azimuth)
 - Capture more subtle parameters like secondary structure, specific nucleotides, chromatin state
 - Indirect measurements of cleavage efficacy!
 - Organism dependency
- gRNA sequence composition does play a role and impacts efficacy across organisms



Tasks (gRNA_design_session1.pdf):

- Visualize possible gRNAs for *galK* KO on sequence, annotate with GC%, efficiency/off-target predictions, location
- Select two gRNAs that you think would work well. Motivate your choice.
- Compare your results to CHOPCHOP or CRISPOR



gRNA design
session 1
NHEJ/whole gene
deletion